

Melanie Angela Zaidel

zaidel.3@osu.edu | [melanieAzaidel.github.io](https://github.com/melanieAzaidel)

 [melaniezaidel7](#) |  [melanieAngelazaidel](#) |  0009-0000-1925-7269

OBJECTIVE

Seeking to be a Faculty member at a college/university, specializing in Astroparticle Physics Theory and Multi-messenger Astronomy, with priorities in teaching/mentoring. While I love studying the universe, my biggest motivator is working and connecting with people, especially students whom I can grow alongside.

EDUCATION

- **Ph.D. in Physics** August 2022 - 2028
Ohio State University Columbus, OH
 - University Fellow, Patrick S. Osmer Fellow
 - National Science Foundation Graduate Research Fellow
 - Advisor: John Beacom
- **M.S. in Physics** August 2022 - July 2024
Ohio State University Columbus, OH
 - Candidacy Topic: *How do Stars Shine? (According to Physicists.)*
- **B.S. in Physics, Astronomy & Astrophysics** June 2018 - May 2022
Pennsylvania State University State College, PA
 - Ronald E. McNair Scholar, Penn State Millennium Scholar
 - Advisor: Kohta Murase
 - Thesis: *Constraining Metastable Very Heavy Dark Matter using the Extragalactic Gamma-Ray Background*

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [J.1] Zaidel, M. A. and Beacom, J. F. (2025). **Probing the conditions of the solar core using ^8B neutrinos**. Phys. Rev. C 112, 055801. Preprint [on arXiv 2504.10583](#).

GRADUATE AWARDS

- **Nomination for the Department of Physics Mentoring Excellence Award** March 2026
Ohio State University 
- **Nomination for the College of Arts and Sciences Graduate Student Award for Mentoring Excellence** March 2026
Ohio State University 
- **Nomination for the Graduate Associate Leadership Award** October 2025
Ohio State University 
- **National Science Foundation Graduate Research Fellowship Award** April 2023
National Science Foundation Division of Graduate Education 

PROJECTS

- **How do Stars Shine (According to Physicists.)** June 2024 - July 2024
Solar Physics 
- **Probing the conditions of the solar core using ^8B neutrinos** September 2023 - April 2025
Solar Neutrino Astrophysics 
- **Constraining Metastable Very Heavy Dark Matter using the Extragalactic γ -Ray Background** June 2021 - July 2022
Dark Matter Astrophysics

TEACHING AND MENTORING

- **Head Teaching Assistant** Summer 2026 - ongoing
OSU Polaris Mentorship Course 
 - Elected by OSU Polaris Leadership as the Head TA to lead the Mentorship Course, a 1-credit course for Freshmen and Sophomore undergrads in Physics/Astronomy
 - Designs interactive course material centered around building community and soft-skills for students who have had less access to physics/astronomy
 - Leads discussions between undergraduate Mentees and graduate student Mentors about impostor syndrome, science communication, and related topics
 - Teaches and manages the course alongside co-TA
- **Organizer** Summer 2024, 2025
Mentoring for Undergraduate Researchers in Summer (MURS)
 - Led a self-designed mini-workshop series for several summer undergraduate research students

- Designed weekly one-hour activities including science communication, keeping a research notebook, forming effective professional relationships with faculty, and Python programming
- Met with students individually to advise them on research tasks, presentations, and career development
- **Workshop Presenter** Summer 2024
OSU Summer Undergraduate Research Program in Astrophysics (SURP) 
 - Designed and presented four one-hour talks to SURP students about the process and experience of applying to graduate school in Physics and Astronomy
 - Answered student questions regarding graduate school applications, fellowships, and timelines
 - Served on a question-and-answer panel with fellow graduate students regarding my path in academia
- **Graduate Student Mentor** August 2023 - Present
OSU Polaris Mentorship Course 
 - Attended training on creating effective mentoring relationships between graduate and undergraduate students
 - Participated in and facilitated mentee-focused in-class discussions on topics including impostor syndrome, science communication, and diverse perspectives in STEM
 - Closely mentored 2 students per year (totaling 6 to date) of the Mentorship Course in psychosocial mentoring, academic advising, professional development, and exploration of topics in physics and astronomy
 - Designed 8-hour self-led research projects for each mentee including interactive lessons, introductions to Python programming, and research poster development
 - Created undergraduate project cookbooks posted on GitHub for future mentors to follow
- **Course Design Institute Attendee** February 2024
OSU Drake Institute for Teaching and Learning 
 - Attended 12 hours of guided workshop about designing effective course curriculum
 - Designed a one-semester course for advanced undergraduates in Astroparticle Physics Research Methods
 - Created scaffolded lesson plans, learning goals, and assessments for the course
- **Academic Facilitator (AF)** April 2023 - October 2023
OSU Polaris, Undergraduate Residential Summer Access (URSA) Program 
 - Designed and presented interactive physics and astronomy-based curriculum centered around the theme of dark matter for a 2-week early-arrival program for incoming undergraduates
 - Mentored 14 students from underrepresented backgrounds in science identity, self-efficacy, and sense of belonging
 - Taught 8 self-designed one-hour classes on fundamental physics, computational resources, scientific reasoning, and dark matter astrophysics
 - Provided teaching support on 8 of fellow AF's classes during lessons
 - Led AF administrative work including planning meeting, scheduling classes, and keeping records on developments of lesson plans and teaching materials
 - Created surveys and collected student feedback on URSA program classes and activities
 - Authored formal report on URSA program design and implementation for use in reporting to department stakeholders

LEADERSHIP AND SERVICE

- **Upward Bound Speaker and Volunteer** June 2025
OSU Department of Astronomy 
 - Volunteered for OSU Department of Astronomy's Upward Bound Astrophysics Institute for Columbus inner-city high-school students
 - Assisted in designing program activities, presentations, and audience participation
 - Presented an accessible and engaging 15-minute talk about the Sun's energy source and how it relates to the solar neutrino problem
- **ASPIRE Speaker and Volunteer** June 2025
OSU Center for Cosmology and AstroParticle Physics (CCAPP) ASPIRE 
 - Volunteered for OSU CCAPP's Achieving in Science through Physics Instrumentation Research and Exploration (ASPIRE) program for high-school students
 - Provided logistical support for program events including inviting speakers, sharing instructional material, and advising on curriculum development
 - Presented an accessible and engaging 15-minute talk about the Sun's energy source and how it relates to the solar neutrino problem
- **Undergraduate Residential Summer Access (URSA) Program Liaison** December 2023 - November 2025
OSU Polaris 
 - Led recruiting, interviewing, and onboarding of URSA graduate student Facilitators

- Collected and documented institutional knowledge, procedures, and best practices regarding the URSA program at-large
- Supported Facilitators in teaching, inter-team communication, conflict resolution, and navigation of OSU organizational structure
- Served as a point-of-contact between URSA Facilitators and Polaris Leadership
- Collected participant feedback on URSA program activities

• Leadership Member

February 2024 - Present

OSU Polaris



- Leads weekly Leadership meetings to discuss progress on Polaris initiatives
- Designs and maintains the Polaris website
- Assists in curriculum development and facilitation of the Polaris Mentorship Course
- Communicates with Physics and Astronomy Departments and College of Arts and Sciences' organizational structures
- Assists in recruiting student participants for Polaris initiatives
- Mentors undergraduates and junior graduate students in leadership within Polaris
- Meets with Department and College administration to secure funding and organizational support for Polaris
- Assists in managing the OSU Polaris GitHub organization and maintaining public repositories

WORKSHOPS

• Your Leadership Legacy Starts Now

February 5, 2026

Columbus, OH

• 4th Santa Cruz School on Multi-messenger Astrophysics

July 2025

Santa Cruz, CA



PRESENTATIONS

• Rising Researchers Seminar Series

February 2026

Remote Invited Talk



• Phenomenology in Indiana, Kentucky, Illinois, Michigan, and Ohio (PIKIMO) 19 Conference

November 2025

Evanston, IL



• CCAPP Seminar Graduate Student Posters

September 2025

Columbus, OH



• Student Talks at Network for Neutrinos, Nuclear Astrophysics, and Symmetries

July 2025

Santa Cruz, CA



• Phenomenology 2025 Symposium

May 2025

Pittsburgh, PA



• 2025 Edward F. Hayes Advanced Research Forum

February 2025

Columbus, OH

• Department of Physics Graduate Student Research Poster Session

September 2023

Columbus, OH

ADDITIONAL INFORMATION

Advanced Graduate Coursework: Numerical and Statistical Methods in Astronomy, Order of Magnitude Astronomy, Learning from Data: Bayesian Methods for Physicists

Programming Languages: Python, Mathematica, LaTeX

Human Languages: English (native fluency), Spanish (native fluency)